Secondary Enucleations for Uveal Melanoma: A 7-Year Retrospective Analysis

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• PURPOSE: To describe the indications for secondary enucleations in uveal melanoma and analyze associations and outcomes.
• DESIGN: Retrospective interventional case series.
• METHODS: Data of patients who underwent secondary enucleation for uveal melanoma in the London Ocular Oncology Service, between 2008 and 2014, were retrieved from medical records analyzed. Cox regression model was performed to analyze associations with secondary enucleation and metastases and Kaplan-Meier estimates to assess the probability of metastatic spread and death.
• RESULTS: During the study period 515 enucleations were performed for uveal melanoma, 99 (19%) of which were secondary enucleations. Tumors were located at the ciliary body in 21 eyes (21%), juxtapapillary in 31 (31%), and choroid elsewhere in 47 (48%). Primary treatment included Ru^{106} plaque radiotherapy, proton beam radiotherapy, and transpupillary thermotherapy in 85, 11, and 3 eyes, respectively. Indications for secondary enucleation were tumor recurrence in 60 (61%), neovascular glaucoma in 21 (21%), and tumor nonresponse in 18 eyes (18%). Twenty patients (20%) were diagnosed with metastasis and 12 out of 20 died of metastatic spread. On multivariate analysis, juxtapapillary tumor location was found to associate with tumor nonresponse ($P = .004$) and nonresponding patients with metastatic spread ($P = .04$).
• CONCLUSIONS: Indications for secondary enucleations for uveal melanoma were tumor recurrence, neovascular glaucoma, and tumor nonresponse. This review identified a possible high-risk group (nonresponse), which proved radioresistant to treatment. These tumors were more frequently found in the juxtapapillary location and were associated with metastatic spread. (Am J Ophthalmol 2015;160(6):1104–1110. © 2015 by Elsevier Inc. All rights reserved.)

UVEAL MELANOMA IS THE MOST COMMON PRIMARY intraocular malignancy in adults, with an incidence of approximately 6 individuals per million population.[1] Three decades ago, the main treatment for uveal melanoma was primary enucleation, aimed at extending survival by removing the eye containing a tumor with metastatic potential. Following the Collaborative Ocular Melanoma Study (COMS) trial, which showed no advantage for enucleation in respect to melanoma-related mortality,[2] treatment now consists mainly of eye-preserving therapies, with plaque brachytherapy currently being the most common.[3] Despite this shift, some patients still require a secondary enucleation after initial conservative treatment, owing to tumor growth or development of neovascular glaucoma (NVG).[4] Reported rates of secondary enucleation for uveal melanoma range widely from 3.9% to 31.2%, according to different studies.[2,4–16] It is noteworthy that most of these reports do not focus on secondary enucleation specifically but rather on treatment outcomes in general. The few cohorts that discuss secondary enucleation are limited in size.[2,4] The objective of this study is to describe a large case series of uveal melanoma patients that underwent secondary enucleation and determine the clinical indications for eye removal and associated tumor risk factors, and to report patient survival.

METHODS

ALL ENUCLEATION CASES FOR UVEAL MELANOMA performed in the London Ocular Oncology Service from January 1, 2008, through December 31, 2014, were retrieved from the surgical log book. Of these, only patients that underwent secondary enucleation were included in the study and their medical records were retrospectively reviewed. The data retrieved from the charts included sex, age, ocular and systemic medical history, laterality, type and timing of primary and salvage treatments, radioactive treatment dosage calculations, surgical notes, indications for and timing of secondary enucleation, clinical examinations throughout the follow-up including early and late postoperative complications, pathology results, development of metastasis, and death from metastatic disease or other causes. Missing medical data were retrieved by contacting the patient's general practitioner (performed by I.D.F.). Tumor location was categorized as originating from the ciliary body; the juxtapapillary choroid, defined as tumor edge located within 1.5 mm from the optic disc (hereafter referred to as juxtapapillary); or choroid elsewhere (hereafter referred to as choroid). The

Accepted for publication Aug 20, 2015.
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0002-9394/$36.00
http://dx.doi.org/10.1016/j.ajo.2015.08.034